

# Interactive Segmentation of Whole-body PET/CT Lesions

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## Abstract

*Lesion segmentation in PET/CT scans is clinically important but time-consuming, and automatic predictions still require manual correction. We present an interactive framework in which an nnU-Net v2 model produces an initial lesion segmentation that a DynUNet-based model then refines locally from user scribbles and the current prediction. To make training and evaluation possible without manual interaction, we introduce a configurable robot user that simulates different annotation strategies. Practical usability is demonstrated through the development of a modular web application whose backend model is interchangeable. The evaluation over five interaction rounds with our refinement stage consistently improves segmentation overlap, while aggregated lesion detection remains stable.*

## 1. Introduction

Lesion segmentation in combined positron emission tomography and computed tomography (PET/CT) scans is an important step in the diagnosis, treatment planning, and monitoring of oncological diseases [10, 13]. Manual annotation is labor intensive and requires considerable expertise [4, 8]. Deep learning can substantially reduce this workload, but automatic predictions still need correction [12], particularly for small or ambiguous lesions.

Interactive segmentation bridges this gap by combining an automatic prediction with sparse user feedback such as clicks or scribbles, which the model incorporates to iteratively refine its output [11, 14]. This keeps experts in control of the final quality, but it also turns latency into a design constraint rather than a free improvement: feedback that does not run at interactive speed will not be used in clinical practice.

We contribute (i) a two-stage interactive model for PET/CT lesion segmentation, in which a DynUNet refinement model [1] updates an automatic nnU-Net prediction [6] inside a crop around each scribble, (ii) a robot user that simulates different annotation behaviors and thereby enables systematic training and evaluation without manual interaction,

and (iii) a modular web application that abstracts the underlying segmentation model and supports interchangeable backends. Our evaluation shows that iterative refinement consistently improves segmentation overlap while aggregated lesion detection stays comparable. Beyond the model itself, the robot user and the application are reusable tools for future research on interactive medical image segmentation.

## 2. Method

Our interactive segmentation framework consists of three main parts, visualized in Fig. 1. Stage 1 predicts an initial lesion segmentation with nnU-Net v2 [6] (Sec. 2.1). Corrective scribbles, drawn either by a user in the web application of Sec. 4 or by the robot user of Sec. 2.2 during training and evaluation, are then passed to the DynUNet-based stage 2 (Sec. 2.3), which refines the prediction. Stage 2 can be flexibly repeated.

### 2.1. Initial Automatic Segmentation Model

nnU-Net derives a dataset fingerprint that captures characteristics such as image spacing and dimensions, and configures preprocessing, network architecture, training, and inference from it, which yields a strong baseline with minimal manual tuning. All scans are resampled to a common voxel spacing, the CT and PET channels are normalized separately, and prediction runs as sliding-window inference over automatically sized patches. We deploy a single trained model (single pass: 70s) instead of the default five-fold ensemble, because the ensemble linearly scales the inference time per examination and thus conflicts with a responsive interactive system.

Given the limited size of the dataset [4], we use a stratified patient-level split. Splitting on the patient level prevents leakage by keeping all scans of a patient in the same subset. We stratify with respect to PET tracer (FDG or PSMA), diagnosis category (lung cancer, lymphoma, melanoma, prostate cancer, or negative), and examination positivity (presence of at least one annotated lesion). Figure 2 shows a representative prediction, in which out of 2 lesions only the larger one was detected. This reveals the remaining potential to improve segmentation performance.

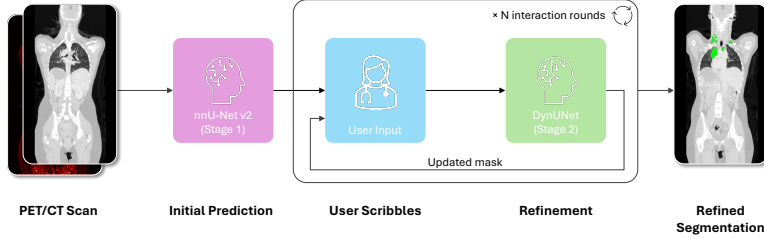


Figure 1. **End-to-end overview.** nnU-Net predicts an initial lesion segmentation from the CT and PET volumes. The user, simulated by the robot user during training and evaluation, draws corrective foreground or background scribbles, and the DynUNet refinement model updates the segmentation iteratively.

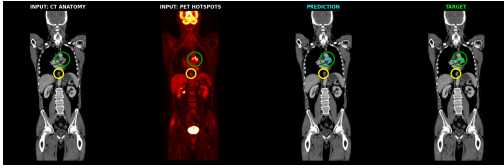


Figure 2. Representative coronal slice of the input and output of the **stage-1 model**. From left to right: CT image, PET image, predicted binary lesion mask, and ground-truth annotation. The green circle marks a correctly detected lesion, the yellow circle a missed one.

## 2.2. Robot User

Training and evaluating the refinement model requires far more corrective scribbles than can be drawn by hand. We therefore introduce a *robot user* that simulates a human annotator by producing one corrective scribble in response to the current segmentation error. It first decomposes the disagreement between prediction and ground truth into per-instance error components, labeling every voxel as true positive (TP), false positive (FP), or false negative (FN). Each component is either a ground-truth lesion that was missed or only partially detected (foreground scribble), or a spurious FP blob (background scribble).

From this decomposition, one interaction is generated in three stages (Fig. 3). *Instance selection* chooses the error component to correct according to a configurable *persona* (e.g. *largest error first*, *biggest lesion first*, *detection completeness*, or *attention to detail*) that defines how components are scored and whether the top-scoring one is taken deterministically or sampled probabilistically by size. *Slice selection* picks the axial slice to draw on, either the slice of maximal error area or an error-area-weighted sample with a temperature parameter controlling slice diversity. *Scribble generation* places the stroke using one of the three generators of the autoPET-V challenge, tracing the contour of the error region (*boundary*), its interior centerline (*skeleton*), or a connection between two random points (*line*). An optional perturbation step inspired by [7, 16] warps, fragments, and thickens the stroke to mimic the variability of hand-drawn input.

For validation and evaluation we fix the random seeds and allocate the persona and generator variations evenly across cases, so that every model configuration is compared consistently. Training instead samples the variations according to the respective configuration, which exposes the model to more diverse annotation behavior.

## 2.3. Interactive Refinement Model

Stage 2 corrects the stage-1 segmentation from the user scribbles and the current prediction. We primarily train DynUNet [1] variants, because its architecture is more readily adaptable for interactive refinement than nnU-Net, whose dedicated interactive extension nnInteractive [7] remains incomplete. For comparison we additionally train a second-stage nnU-Net. The two models differ in how the scribbles are supplied and in the spatial scope of a prediction.

**DynUNet.** Building on the idea that interactive refinement should act only within the region of the interaction rather than the full volume [3], we restrict each prediction to a fixed  $128^3$  voxel crop centered on the current scribble. The five input channels concatenate the CT and PET modalities, the prior mask (inspired by [2, 7]), and the guidance signal (foreground & background scribble) encoded as Gaussian heatmaps. The prior mask is the current segmentation, which is the stage-1 output in the first round and the previous round’s output afterwards. The network maps the  $B \times 5 \times 128^3$  input to a single-channel sigmoid output of the same spatial size, the updated lesion mask for the crop. We optimize a combined Dice and cross-entropy loss with Adam (learning rate  $10^{-4}$ ) under a cosine-annealing schedule.

**Single-step training.** Every training sample performs a single refinement step from the fixed stage-1 segmentation rather than from a rolled-out interaction sequence. Longer rollouts proved unstable. Training from scratch without pre-trained weights, the network initially degraded the already accurate stage-1 segmentation into many spurious connected components before learning to use the guidance, which also slowed the CPU-bound robot user, whose error analysis cost grows with the number of components. Starting every step from the same stable baseline lets the model first learn a

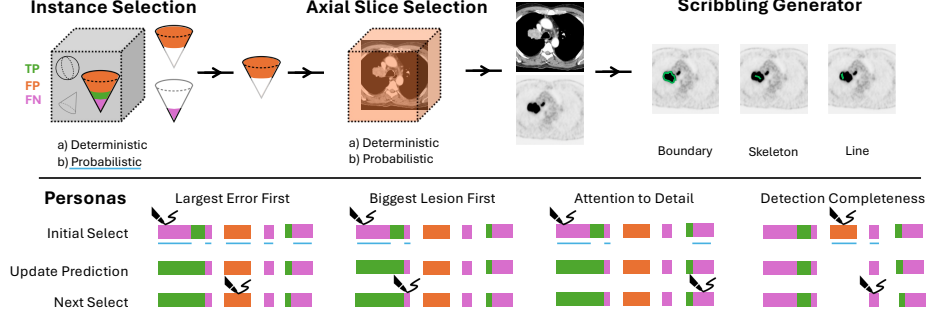


Figure 3. The **robot user** simulates an annotator in three stages: it selects an error component (instance selection), an axial slice, and draws a scribble with one of three generators. The error decomposition marks each voxel as TP (green), FP (orange), or FN (magenta). The personas below illustrate how different targeting policies choose the next correction and how the prediction might update in response.

reliable one-step correction. It also makes the per-case error decomposition identical across epochs, so it is cached once and each epoch only re-runs the lightweight scribble sampling and reconstructs the masks for the scribble generator. At evaluation the same one-step model is applied iteratively, writing its refined output back into the full volume from which the next round determines its crop.

**Second-stage nnU-Net.** The alternative refinement model receives the CT and PET volumes together with the two guidance signal channels, but no prior mask. Its scribbles are generated once and reused every epoch instead of being sampled dynamically. It also performs sliding-window inference over the full scan, so a single interaction can update the entire volume instead of a local crop.

### 3. Results

**Protocol.** All numbers below follow the autoPET-V metric definitions. Overlap is reported as the mean per-case Dice similarity coefficient (DSC) over the tumor-positive cases, for which DSC is defined. Detection is reported as the lesion-level F1 pooled over all cases, where a ground-truth lesion counts as detected at an intersection over union of at least 0.1. Because the F1 is pooled, a tumor-negative case can only ever contribute false positives, so we additionally report the mean false-positive volume (FPV) on that cohort. Every run starts from the same stage-1 nnU-Net prediction (round 0) and is refined for five rounds, one robot user scribble each. Model and interaction choices were made on the validation split; the test split was scored once, at the end.

**Local refinement as a possible safety guard.** Table 1 compares the crop-restricted DynUNet with the second-stage nnU-Net, which re-predicts the whole volume. Their overlap after five rounds is nearly identical (0.694 vs. 0.684 DSC), but the DynUNet realizes two thirds of its gain already in the first round (+0.068 DSC), whereas the second-stage model gains almost nothing from the first scribble (+0.004 DSC) and ends with a pooled F1 and a false-positive burden below the baseline it started from. A plausible explanation is the

| Refinement model       | DSC <sub>5</sub> | $\Delta$ DSC | $\overline{\text{DSC}}$ | F1 <sub>5</sub> | FPV <sub>5</sub> |
|------------------------|------------------|--------------|-------------------------|-----------------|------------------|
| none (stage 1)         | 0.591            | —            | —                       | <b>0.681</b>    | 52.1             |
| nnU-Net+scribbles      | 0.684            | 0.094        | 0.645                   | 0.657           | 59.9             |
| DynUNet, jitter+priors | <b>0.694</b>     | <b>0.104</b> | <b>0.674</b>            | 0.680           | <b>43.9</b>      |

Table 1. **Refinement models on the validation split** (231 examinations, 151 tumor-positive). Subscript 5 denotes the value after five rounds;  $\overline{\text{DSC}}$  is the mean across all six rounds, which rewards reaching a good segmentation early. FPV is the mean false-positive volume in mL on the 80 tumor-negative cases.

spatial scope. Rewriting the whole scan lets one scribble also disturb regions that were already correct, which a crop-restricted update cannot do. The two models also differ in backbone and in how the scribbles are supplied, so this remains a hypothesis rather than an isolated cause.

**Test set and interaction strategy.** Figure 4 reports the DynUNet configuration on the 239 unseen test examinations. Under the realistic balanced protocol the mean DSC rises from 0.617 to 0.696, with the bulk of the gain again in the first round. Under the greedy *largest error first* annotator, which upper-bounds what five scribbles can achieve, it reaches 0.744 and is still climbing at round five. Detection stays flat throughout, as the pooled F1 remains within 0.01 of its baseline of 0.708 under either protocol. The gap between the two protocols is the practically relevant caveat: more than a third of the achievable gain depends on where the user chooses to correct, not on the model.

**Small lesions remain challenging.** Stratified by lesion volume, the interaction pays off almost entirely on lesions that are already large enough to be found. Over five rounds, recall on the test set under the balanced protocol improves from 0.906 to 0.945 above 20 mL and from 0.767 to 0.796 in the 5–20 mL bucket, but it decreases slightly below 1 mL (0.656 to 0.627), a bucket that holds about 44% of all annotated lesions. A single scribble on the largest remaining error is a poor instrument for recovering many small, scat-

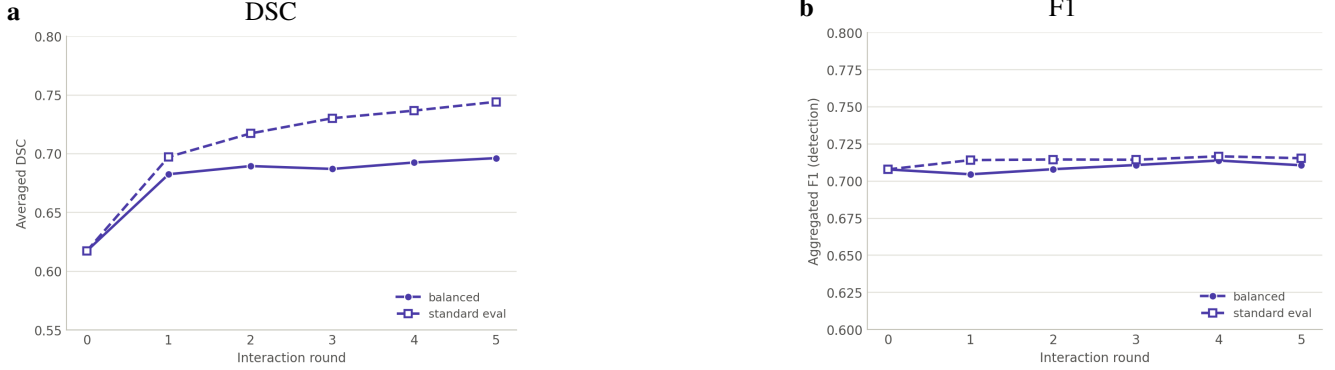


Figure 4. **End-to-end evaluation results on test split** (239 examinations), with the stage-1 prediction at round 0 and five refinement rounds. **a**, mean DSC over the 157 tumor-positive cases increases over the rounds. **b**, lesion F1 pooled over all cases moves sideways.

tered lesions, which may also explain why the pooled F1, weighting every lesion equally, barely moves while the DSC improves.

#### 4. Interactive Segmentation Tool

We further developed a web application for interactive lesion segmentation, aimed at computer vision researchers who want to try out and compare approaches or annotate future datasets. It supports the loading of volumetric medical scans, the drawing of user scribbles for segmentation guidance, and the plug-and-play execution of different segmentation models.



Figure 5. User interface of the segmentation tool, with a CT and a PET scan loaded and rendered alongside the ground-truth mask and the prediction of the previous iteration. The pen tool draws a foreground scribble on an undersegmented area.

**Architecture.** The frontend renders the volumetric data in real time with the WebGL-based NiiVue library [5]. It is backed by a Python backend, which offers a REST API, converts the received data appropriately, and runs the actual segmentation model inside a Docker container [9].

**Abstraction from the segmentation model.** Containerization allows models that share an input and output interface to be swapped without friction, *e.g.* for different approaches to the same challenge. On top of it, a small abstraction layer maps any available data into and out of the container according to a configuration file that describes the container’s inputs and outputs. Models with a different interface only

require a new configuration file, which can then be selected for all matching models.

**Handling statefulness.** Iterative refinement is inherently stateful. We therefore provide each model container with a directory that is kept intact between iterations. The lifecycle of these directories is tied to case identifiers assigned by the frontend, which uses the input modalities of the current model to distinguish independent cases.

#### 5. Conclusion and Outlook

We presented an interactive framework for PET/CT lesion segmentation that refines an nnU-Net baseline from user scribbles, together with a configurable robot user and a modular web application with an exchangeable backend model. Restricting each update to a local crop consistently improved segmentation overlap over successive interaction rounds while keeping lesion detection stable, and it avoided the degradation of correctly segmented regions that we observed when scribbles were integrated globally into the second-stage nnU-Net. Small lesions remained the hardest case, and a substantial share of the attainable gain depended on where the user corrects rather than on the model. Promising directions include adding anatomical context (*e.g.* organ maps from TotalSegmentator [15]), using the robot user to study alternative interaction strategies, and evaluating the web application with clinical users.

#### References

- [1] M. Jorge Cardoso et al. Monai: An open-source framework for deep learning in healthcare. *arXiv preprint arXiv:2211.02701*, 2022. 1, 2
- [2] Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Didac Suris, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, Jie Lei, Tengyu Ma, Baishan Guo, Arpit Kalla, Markus Marks, Joseph Greer, Meng Wang, Peize Sun, Roman Rädle, Triantafyllos Afouras, Effrosyni Mavroudi, Katherine Xu,

- Tsung-Han Wu, Yu Zhou, Liliane Momeni, Rishi Hazra, Shuangrui Ding, Sagar Vaze, Francois Porcher, Feng Li, Siyuan Li, Aishwarya Kamath, Ho Kei Cheng, Piotr Dollár, Nikhila Ravi, Kate Saenko, Pengchuan Zhang, and Christoph Feichtenhofer. Sam 3: Segment anything with concepts, 2025. [2](#)
- [3] Xi Chen, Zhiyan Zhao, Yilei Zhang, Manni Duan, Donglian Qi, and Hengshuang Zhao. Focalclick: Towards practical interactive image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1300–1309, 2022. [2](#)
- [4] Sergios Gatidis, Tobias Hepp, Marcel Früh, Christian La Fougère, Konstantin Nikolaou, Christina Pfannenberger, Bernhard Schölkopf, Thomas Küstner, Clemens Cyran, and Daniel Rubin. A whole-body fdg-pet/ct dataset with manually annotated tumor lesions. *Scientific Data*, 9(1):601, 2022. [1](#)
- [5] Taylor Hanayik, Chris Rorden, Christopher Drake, Jens Ochsmeier, Matt McCormick, Alexis, Kabilar Gunalan, Anthony Androulakis, Paul Wighton, John Lee, Paul Taylor, Edwin Bennink, Shun, Nell Hardcastle, Yaroslav Halchenko, Florian Rupprecht, Korbinian Eckstein, Guilherme Povala, Paul McCarthy, Sumit Jha, Albert Armea, Chris Markiewicz, Claude, Ilias Mahboub, Jan-Hendrik Müller, Pierre, Rahul Chaudhary, Ravi Ranjan, and Adam Basha. niivue/niivue: @niivue/niivue-v0.68.2, 2026. [4](#)
- [6] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021. [1](#)
- [7] Fabian Isensee, Maximilian Rokuss, Lars Krämer, Stefan Dinkelacker, Ashis Ravindran, Florian Stritzke, Benjamin Hamm, Tassilo Wald, Moritz Langenberg, Constantin Ulrich, Jonathan Deissler, Ralf Floca, and Klaus Maier-Hein. nninteractive: Redefining 3d promptable segmentation, 2025. [2](#)
- [8] Zdravko Marinov, Paul F. Jäger, Jan Egger, Jens Kleesiek, and Rainer Stiefelhagen. Deep interactive segmentation of medical images: A systematic review and taxonomy, 2024. [1](#)
- [9] Dirk Merkel. Docker: lightweight linux containers for consistent development and deployment. *Linux Journal*, 2014(239): 2, 2014. [4](#)
- [10] Maximilian Rokuss, Balint Kovacs, Yannick Kirchhoff, Shuhan Xiao, Constantin Ulrich, Klaus H. Maier-Hein, and Fabian Isensee. From fdg to psma: A hitchhiker’s guide to multitracer, multicenter lesion segmentation in pet/ct imaging, 2024. [1](#)
- [11] Konstantin Sofiuk, Ilia Petrov, and Anton Konushin. Reviving iterative training with mask guidance for interactive segmentation. *arXiv preprint arXiv:2102.06583*, 2021. [1](#)
- [12] Michael J Trimpl, Sergey Primakov, Philippe Lambin, Eleanor P J Stride, Katherine A Vallis, and Mark J Gooding. Beyond automatic medical image segmentation—the spectrum between fully manual and fully automatic delineation. *Physics in Medicine Biology*, 67(12):12TR01, 2022. [1](#)
- [13] Richard L. Wahl, Heather Jacene, Yvette Kasamon, and Martin A. Lodge. From recist to perclist: Evolving considerations for pet response criteria in solid tumors. *Journal of Nuclear Medicine*, 50(Suppl 1):122S–150S, 2009. [1](#)
- [14] Guotai Wang, Maria A. Zuluaga, Wenqi Li, Rosalind Pratt, Premal A. Patel, Michael Aertsen, Tom Doel, Anna L. David, Jan Deprest, Sebastien Ourselin, and Tom Vercauteren. Deepi-geos: A deep interactive geodesic framework for medical image segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(7):1559–1572, 2019. [1](#)
- [15] Jakob Wasserthal, Manuel Meyer, Hans-Christoph Breit, Julie Cyriac, Shan Yang, and Martin Segeroth. Totalsegmentator: Robust segmentation of 104 anatomical structures in ct images. *Radiology: Artificial Intelligence*, 5(5), 2023. [4](#)
- [16] Hallee E. Wong, Marianne Rakic, John Gutttag, and Adrian V. Dalca. Scribbleprompt: Fast and flexible interactive segmentation for any biomedical image. *European Conference on Computer Vision (ECCV)*, 2024. [2](#)